SPARC: Spatial-Aware Path Planning via Attentive Robot Communication

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***Abstract*—Efficient communication is critical for decentralized Multi-Robot Path Planning (MRPP), yet existing learned communication methods treat all neighboring robots equally regardless of their spatial proximity, leading to diluted attention in congested regions where coordination matters most. We propose Relation-enhanced Multi-Head Attention (RMHA), a communication mechanism that explicitly embeds pairwise Manhattan distances into the attention weight computation, enabling each robot to dynamically prioritize messages from spatially relevant neighbors. Combined with a distance-constrained attention mask and GRU-gated message fusion, RMHA integrates seamlessly with MAPPO for stable end-to-end training. In zero-shot generalization from 8 training robots to 128 test robots on 40×40 grids, RMHA achieves approximately 75% success rate at 30% obstacle density—outperforming the best baseline by over 25 percentage points. Ablation studies confirm that distance-relation encoding is the key contributor to success rate improvement in high-density environments.**

***Index Terms*—Multi-robot path planning, graph attention mechanism, multi-head attention, communication optimization, cooperative decision-making**

I. Introduction and Related Work

Multi-Robot Path Planning (MRPP) is a core problem in artificial intelligence and robotics, aiming to design collision-free paths for multiple robots to navigate from their respective starting positions to goal positions in a known environment [1]. In this paper, the terms *agent* and *robot* are used interchangeably. In the standard MRPP formulation, a graph structure represents the robots' workspace together with sets of start and goal vertices. At each time step, a robot may either stay at its current vertex or move to an adjacent one [2]. A solution to an MRPP instance is a set of paths—one per robot—that are free of both vertex and edge collisions. Finding such a solution is known to be NP-hard.

An MRPP instance is defined by a graph, a set of robots, and their respective start–goal pairs. The problem can be viewed as a planning challenge whose objective is to find a collision-free path set (the *solution*) for a given instance. An MRPP *algorithm* (or *solver*) refers to the method used to compute such a solution.

MRPP has broad applications in intelligent warehousing systems [3], urban traffic networks [4], autonomous vehicles, and electronic games, among others [5]. As a foundational task in building physical multi-robot systems, MRPP plays a pivotal role in the design phase and its importance is expected to grow as demand for cooperative and competitive multi-robot scenarios increases.

In the warehouse logistics domain in particular, the rapid expansion of e-commerce has made intelligent logistics systems indispensable. Autonomous Mobile Robots (AMR) [6] are increasingly deployed in unmanned warehouses. For such AMR-based warehousing systems, path planning is the core scheduling component. However, as the number of robots and the complexity of tasks increase, efficiently and safely planning collision-free paths remains a critical challenge.

A. Centralized Planning Methods

Centralized approaches rely on a central controller with complete global information to compute (near-)optimal paths.

A* and its variants. A* is a classic heuristic search algorithm whose evaluation function is $f(n) = g(n) + h(n)$. Extensions to multi-robot settings include LRA*, which adjusts trajectories upon detecting potential conflicts; CA*, which introduces a three-dimensional reservation table to avoid collisions [7]; and HCA*, which employs hierarchical abstraction to improve scalability. However, frequent replanning in dense scenarios leads to high computational overhead.

Conflict-Based Search (CBS) uses a two-level search: the high level discovers and decomposes conflicts, while the low level computes optimal single-robot paths under constraints [8]. A binary constraint tree is constructed until a conflict-free solution is found.

ODrM* combines decoupled and coupled strategies: each robot initially follows its individual optimal path; upon collision, only the conflicting robots are coupled for joint search [9], [10]. While effective as a bounded sub-optimal planner, ODrM* suffers from exponential complexity growth when the number of agents or environment complexity increases.

B. Decentralized Learning-Based Methods

Centralized methods struggle with real-time requirements and scalability as the number of robots and environment size grow. Multi-Agent Reinforcement Learning (MARL) offers an effective alternative under partial observability: agents access only local observations and make fast, decentralized decisions [11].

Learning-based methods typically train a policy that maps local observations (*e.g.*, a limited Field of View, FOV) to discrete actions (up/down/left/right/wait) on a grid world. Major paradigms include Imitation Learning (IL), which imitates demonstrations from centralized solvers, and Reinforcement Learning (RL), which maximizes expected return. In MRPP, the problem is commonly modeled as a Dec-POMDP [11]. PRIMAL [12] integrates physical simulation with hybrid offline–online training; LNS2+RL [14] fuses large

neighborhood search with MARL; Li *et al.* first introduced GNNs for MRPP, encoding FOV interactions and supporting decentralized decisions [15], [16].

C. Motivation and Challenges

Despite progress in learned communication for MRPP, two key challenges remain. **(C1) Spatially-agnostic communication:** Existing methods such as MAGAT [16] and SCRIMP [1] compute attention weights purely from message content, treating a distant robot identically to an adjacent one. In congested regions, this dilutes attention across irrelevant neighbors and degrades coordination. **(C2) Training instability at scale:** Transformer-based communication introduces high-variance gradients in online MARL, causing training collapse when naively applied to large robot populations. Our work addresses both challenges by embedding spatial relations directly into the attention mechanism and constraining communication to local neighborhoods.

D. Contributions

The main contributions of this work are as follows:

1) We propose a **spatial relation-enhanced multi-head attention communication mechanism** that incorporates inter-robot relative distance as edge information into the attention weight computation, enabling communication weights to adapt dynamically to topological relationships and improving cooperative decision-making.
2) We construct an **end-to-end cooperative learning framework based on MAPPO**, introducing distance-constrained local communication and attention masking to reduce communication overhead while maintaining effective information exchange and training stability.
3) We design **communication ablation studies and multi-density scenario evaluations** that quantitatively verify the critical role of distance-relation encoding for success rate improvement and congestion/conflict mitigation, demonstrating stronger robustness in high-density environments.

II. Spatial Relation-Enhanced Multi-Head Attention Communication for Multi-Robot Path Planning

A. System Model and Problem Formulation

1) MRPP System Model: The MRPP problem can be formally defined as a four-tuple $\langle \mathcal{A}, G, \mathcal{O}_{\text{dyn}}, \mathcal{O}_{\text{static}} \rangle$, where: $\mathcal{A} = \{A_1, A_2, \ldots, A_N\}$ is the set of N robots; $G = (V, E)$ is a graph representing the environment with vertex set V (locations) and edge set E (traversable connections); $\mathcal{O}_{\text{dyn}} = \{O_1^d, \ldots, O_M^d\}$ denotes M dynamic obstacles whose positions change over time; and $\mathcal{O}_{\text{static}} = \{O_1^s, \ldots, O_K^s\}$ denotes K static obstacles.

Each robot $A_n \in \mathcal{A}$ is represented by a tuple $\langle s_n, g_n \rangle$, where s_n and g_n are the start and goal positions, respectively. A valid MRPP solution must satisfy the constraints illustrated in Fig. 1:

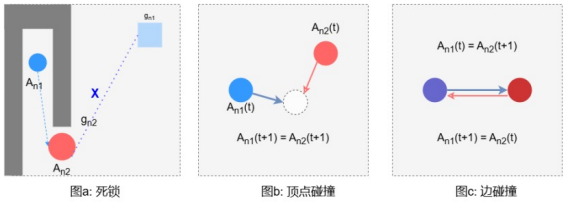


Fig. 1. Illustration of MRPP constraints: vertex collisions, edge collisions, and swap collisions must all be avoided.

1) **Deadlock-free constraint:** There exists a finite time step $T \in \mathbb{Z}^+$ such that all robots reach their goals.
2) **Vertex collision-free constraint:** For any two robots A_n and $A_{n'}$ and any time step $0 \leq t \leq T$, their positions must not coincide, nor may any robot occupy a position held by a static or dynamic obstacle.
3) **Edge collision-free constraint:** No two robots may simultaneously traverse the same edge in opposite directions.

When multiple valid solutions exist, MRPP can be cast as an optimization problem. Common objectives include the *makespan*:

$$M(\pi) = \max_{1 \leq k \leq N} |\Pi_k|, \tag{1}$$

and the *Sum of Costs (SOC)*:

$$\text{SOC}(\pi) = \sum_{k=1}^{N} |\Pi_k|, \tag{2}$$

where $|\Pi_k|$ denotes the path length of robot k. These two objectives optimize the solution from the perspectives of time and total path length, respectively.

Reward design in RL-based MRPP. Table I summarizes the reward mechanisms used in representative RL-based MRPP methods. Core elements include movement penalties and collision penalties to constrain redundant actions and conflict risks, heuristic-based sub-goal positive feedback to guide progressive goal achievement, and mechanisms that suppress blocking, deadlock, and path deviation to improve cooperation efficiency.

2) Problem Formulation: In this work, the MRPP problem is defined by a set of robots $\{1, \ldots, n\}$ and represented as a tuple $\langle G, S, T \rangle$, where $G = (V, E)$ is an undirected graph, $S = \{s_1, \ldots, s_n\} \subset V$ and $T = \{t_1, \ldots, t_n\} \subset V$ are the start and goal positions, respectively [13]. Time is discretized into steps; at each step a robot may move to an adjacent vertex or stay, provided the action does not cause a vertex, edge, or swap collision. The objective is to find the shortest joint collision-free path from S to T.

In the MARL setting, the problem is modeled as a Markov game $\langle \mathcal{N}, \mathcal{O}, \mathcal{A}, R, P, \gamma \rangle$, where $\mathcal{N} = \{1, \ldots, N\}$ is the robot set; $\mathcal{O} = \prod_{i=1}^{N} \mathcal{O}_i$ is the joint observation space; $\mathcal{A} = \prod_{i=1}^{N} \mathcal{A}_i$ is the joint action space; $R : \mathcal{O} \times \mathcal{A} \to [-R_{\max}, R_{\max}]$ is the joint reward function; $P : \mathcal{O} \times \mathcal{A} \times \mathcal{O} \to \mathbb{R}$ is the transition probability; and $\gamma \in (0, 1)$ is the discount factor.

TABLE I
COMPARISON OF REWARD DESIGNS IN RL-BASED MRPP METHODS

Method	Year	Move	Wait[a]	Wait[b]	Coll.	Ego-goal	All-goal	Block	Oscil.	Deviate
PRIMAL	2019	↓	↓	→	↓	–	↑	↓	–	–
GNN	2020	–	–	–	–	–	–	–	–	–
MAPPER	2020	↓	↓	↓	↓	↑	–	–	↓	↓
G2RL	2020	↓	–	–	↓	–	↑	–	–	–
PRIMAL2	2021	↓	↓	→	↓	↑	–	–	–	–
MAGAT	2021	–	–	–	–	–	–	–	–	–
DHC	2021	↓	↓	→	↓	↑	–	↓	–	–
PICO	2022	↓	↓	→	↓	–	↑	–	–	–
DCC	2022	↓	↓	→	↓	↑	–	–	–	–
SACHA	2023	↓	↓	→	↓	↑	–	↓	–	–
SCRIMP	2023	↓	↓	→	↓	→	↑	–	–	–

[a]Not at goal. [b]At goal. ↓ = penalty; ↑ = reward; → = neutral; – = not used.

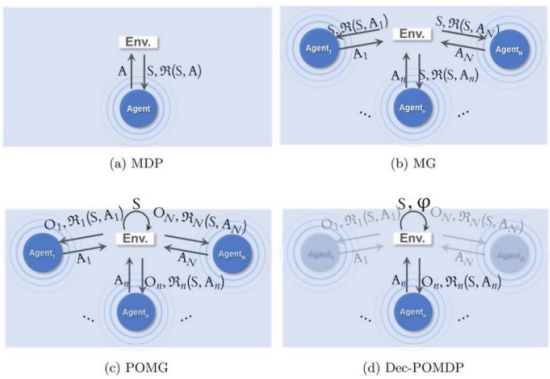


Fig. 2. MARL modeling paradigms for MRPP: MG, POMG, and Dec-POMDP.

At each time step t, robot i receives observation $o_i^t \in \mathcal{O}_i$ and selects action a_i^t according to its policy π_i. The objective is to maximize the cumulative discounted reward:

$$R_\gamma = \sum_{t=0}^{\infty} \gamma^t \, R(o^t, a^t). \tag{3}$$

B. Spatial Relation-Enhanced Multi-Head Attention Communication Algorithm

1) MDP Modeling for MRPP: As illustrated in Fig. 2, MRPP can be modeled as a Markov Game (MG), an extension of the Markov Decision Process (MDP) to multi-agent settings [11]. An MDP is defined by a five-tuple $\langle S, A, T, R, \gamma \rangle$. In an MG with N robots, the action space extends to the joint space $\mathcal{A}^{1\ldots N} = \mathcal{A}_1 \times \cdots \times \mathcal{A}_N$, and the reward function becomes a set $\mathcal{R} = \{R_1, \ldots, R_N\}$ with $R_n : S \times \mathcal{A}_n \to \mathbb{R}$. Under partial observability, the model becomes a Partially Observable Markov Game (POMG) [17], where each robot observes only a local view $o_n \in \mathcal{O}_n$. Alternatively, a Dec-POMDP formulation can be used, represented as $\langle \mathcal{O}, S, \mathcal{A}, T, R, \phi, \gamma \rangle$, where ϕ denotes the conditional observation distribution [18].

To cast MRPP as an RL problem, we construct a discrete two-dimensional grid world. Robots, goals, and obstacles each occupy one cell. At the start of each episode, n unique start and goal positions are randomly assigned within the same connected region. Robots move simultaneously and collisions are checked on the joint action vector. Each robot's FOV is limited to 3×3.

TABLE II
REWARD SETTINGS

Action	Reward
Move (up/down/left/right)	-0.3
Stay (at goal / off goal)	0.0 / -0.3
Collision	-2.0
Blocking	-1.0

State space $\mathcal{S}$: Each robot's observation consists of two parts. The first comprises eight binary matrices: four heuristic maps (one per movement direction, with a cell marked 1 iff executing that action brings the robot closer to its goal) and four maps encoding nearby obstacles, other robots' positions, the robot's own goal, and projected goal positions of observable peers. The second is a 7-dimensional vector containing the normalized distance to the goal (d_x, d_y, d), the extrinsic reward r_{t-1}^e, the intrinsic reward r_{t-1}^i, the minimum Euclidean distance $d_{\min}^{t-1}$ to positions stored in a short-term memory buffer, and the previous action a_{t-1}.

Action space $\mathcal{A}$: $\mathcal{A} = \{\text{Up}, \text{Down}, \text{Left}, \text{Right}, \text{Stay}\}$. Actions are feasible only if they do not cause vertex, edge, or swap collisions.

Reward R: The reward structure is summarized in Table II. Robots that have not yet reached their goals receive a per-step penalty to encourage faster completion. An episode terminates when all robots reach their goals or a maximum of 256 time steps is reached. Communication messages are subject to a one-step delay and are not blocked by obstacles.

2) The RMHA Communication Mechanism: The architecture of the proposed RMHA module is shown in Fig. 3. To design an efficient communication paradigm for MARL-based MRPP, we adopt a communication Transformer [19]–[21] inspired by the graph modeling paradigm. We apply the Transformer Encoder architecture to map the input observation sequence $(o_1, \ldots, o_N)$ to the output action se-

quence $(a_1, \ldots, a_N)$. Each encoder block consists of a spatial relation-enhanced attention mechanism, a multi-layer perceptron (MLP) [22]–[24], and residual connections.

In the standard multi-head attention, the attention score between elements o_i and o_j is computed as the dot product of their query and key vectors:

$$s_{ij} = f(o_i, o_j) = o_i\, W_q^\top W_k\, o_j, \tag{4}$$

which can be viewed as implicit edge information associated with the directed edge $e_{j\to i}$.

We propose a **Spatial Relation-Enhanced Multi-Head Attention** (RMHA) mechanism that explicitly incorporates inter-robot spatial relationships into the attention computation [25]–[27]. Specifically, the Manhattan distance between robots is embedded into a high-dimensional vector space and combined with local observations to jointly compute attention scores:

$$s_{ij} = g(o_i, o_j, d_{i\to j}, d_{j\to i}) = (o_i + d_{i\to j})\, W_q^\top W_k\, (o_j + d_{j\to i}), \tag{5}$$

where $d_{*\to *}$ denotes the embedding vector of the Manhattan distance between the corresponding robots. We choose Manhattan distance over Euclidean distance because robots move on a grid with four cardinal directions, making Manhattan distance the exact shortest-path metric in obstacle-free regions and a tighter lower bound in the presence of obstacles. The encoded observations $(o_1, \ldots, o_N)$ thus capture not only individual robot information but also higher-level spatial relationships through communication.

Communication range constraint. In practice, each robot can only communicate with a limited number of peers due to bandwidth and medium-access contention. To model this, each robot communicates only with those within a specified radius R:

$$\mathcal{N}_i(t) = \{j \mid d_{ij}(t) \le R,\ j \ne i\}, \tag{6}$$

where $d_{ij}(t)$ is the distance between robots i and j at time t. An adjacency-based mask is applied to the attention scores:

$$\tilde{s}_{ij} = \begin{cases} s_{ij}, & \text{if } e_{j\to i} = 1, \\ -\infty, & \text{if } e_{j\to i} = 0, \end{cases} \tag{7}$$

ensuring that only information from connected robots is accessible.

3) Integration with MAPPO: We adopt Multi-Agent Proximal Policy Optimization (MAPPO) [28]–[30] as the base training framework. The neural network model integrates three components (Fig. 4):

(1) Multi-modal observation encoder. A backbone of seven convolutional layers and two max-pooling layers extracts spatial features, followed by three fully connected layers for dimensionality reduction. For robot i at time t, the local observation $o_i^t \in \mathbb{R}^{4\times m\times m}$ is processed by:

$$\hat{o}_i^t = \mathrm{Conv}_2\big(\mathrm{MP}\big(\mathrm{Conv}_1(o_i^t)\big)\big), \tag{8}$$

$$\bar{o}_i^t = \mathrm{FC}_1\big(\mathrm{concat}\big(\hat{o}_i^t,\ \mathrm{FC}_0(v_i^t)\big)\big), \tag{9}$$

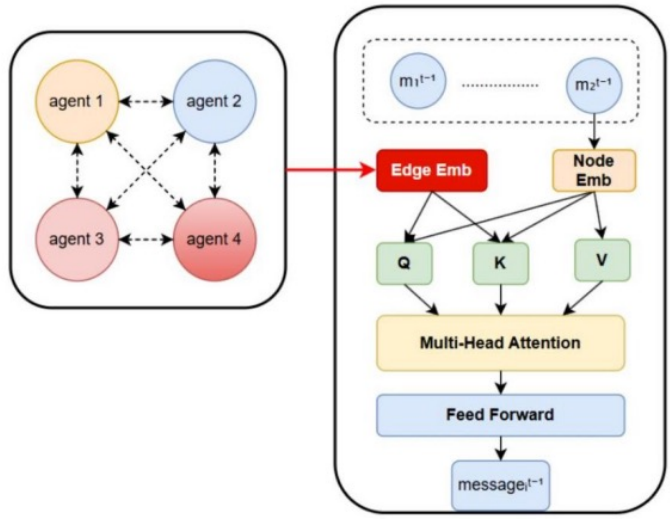


Fig. 3. Architecture of the RMHA communication module.

where v_i^t is the 7-dimensional state vector. Temporal fusion is achieved via an LSTM unit:

$$h_i^t = \mathrm{LSTM}(\bar{o}_i^t, h_i^{t-1}). \tag{10}$$

(2) Distributed communication module. Built on the Transformer encoder, this module deeply fuses the quantized Manhattan distance representation with attention weight computation. For a group of n robots, the queries, keys, and values are:

$$Q = W_q\big(M^{t-1} + W_1 \cdot \mathrm{Distance}\big), \tag{11}$$

$$K = W_k\big(M^{t-1} + W_2 \cdot \mathrm{Distance}\big), \tag{12}$$

$$V = W_v\, M^{t-1}, \tag{13}$$

where $M^{t-1} \in \mathbb{R}^{n\times d}$ is the message matrix from the previous time step, and $W_1 \cdot \mathrm{Distance}$ encodes explicit edge information. The attention weights are computed using scaled dot-product attention:

$$\alpha_i = \mathrm{softmax}\left(\frac{q_i\, K^\top}{\sqrt{d_k}}\right). \tag{14}$$

Information aggregation is performed through multi-head attention:

$$\mathrm{head}_i = \alpha_i\, V, \tag{15}$$

$$m_i^{t-1} = \mathrm{concat}\big(\mathrm{head}_i^1, \ldots, \mathrm{head}_i^h\big)\, W_o. \tag{16}$$

Local communication is enforced via a mask matrix $\mathcal{M}_{ij} = -\infty \cdot \mathbf{1}(\|p_i - p_j\|_2 > r)$, where r is the communication radius.

(3) Output heads. The LSTM hidden state and the communication message are concatenated and fed to four output heads that produce: (i) the message vector for the next time step, (ii) state-value estimates for extrinsic and intrinsic rewards, (iii) the policy distribution $\pi_i(a^t \mid o^t)$, and (iv) a blocking flag indicating whether the robot obstructs other agents' paths.

The policy is optimized using the clipped PPO objective:

$$L_\pi(\theta) = \mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1\pm\epsilon)\hat{A}_t\big)\Big], \tag{17}$$

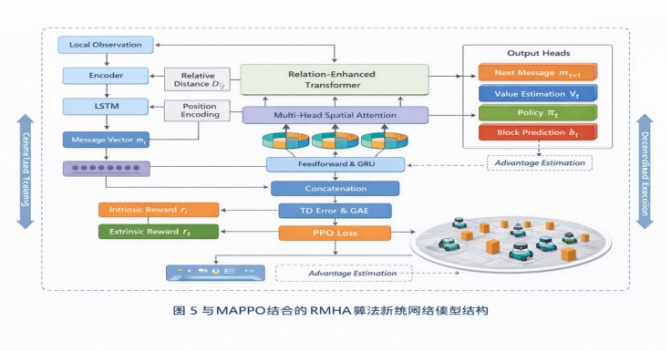


Fig. 4. Network architecture of RMHA integrated with MAPPO.

TABLE III
EXPERIMENTAL PARAMETERS

Parameter	Value
Training robot count	8
Action count	5
Max episode length	256
FOV size	3
World size	$(10, 40)$
Obstacle probability	$(0.0, 0.5)$
Mask distance	40
Move cost	-0.3
Idle cost	-0.3
Goal reward	0.0
Collision cost	-2.0
Blocking cost	-1.0

where $\hat{A}_t$ is the Generalized Advantage Estimation (GAE) and $r_t(\theta) = \pi_\theta(a_t|o_t)/\pi_{\theta_{\text{old}}}(a_t|o_t)$ is the importance sampling ratio.

MAPPO employs Centralized Training with Decentralized Execution (CTDE): during training, agents share environment states, actions, and rewards to build a globally informed joint policy model; during execution, each agent relies solely on local observations for independent decision-making.

The complete training procedure is presented in Algorithm 1.

Algorithm 1 RMHA Training Procedure

Input: Robot count N, episodes K, steps per episode T, FOV size F, Transformer layers L, discount γ, PPO clip ϵ, intrinsic reward params τ, φ, β, M

Initialize encoder params ϕ, Transformer params ψ, output head params θ, replay buffer $\mathcal{B}$
for $k = 0, 1, \ldots, K-1$ **do**
  for $t = 0, 1, \ldots, T-1$ **do**
    Collect observations $o_1^t, \ldots, o_N^t$
    // Encode observations
    for each robot i **do**
      Encode o_i^t via ϕ (Eqs. 8–10)
    end for
    // Transformer-based communication
    for each robot i **do**
      Generate message m_i^t; compute distances D_i^t
    end for
    Add sinusoidal positional embeddings (robot IDs)
    for $l = 1, \ldots, L$ **do**
      Compute multi-head self-attention with distance (Eqs. 11–16)
      Update messages via GRU and FFN
    end for
    // Output heads
    for each robot i **do**
      Concatenate Transformer output with LSTM state
      Generate π_i^t, v_i^t, blocking prediction b_i^t
    end for
    // Value-based conflict resolution
    for each conflicting pair (i, j) **do**
      Compute priority via softmax over value diffs
      High-priority robot keeps action; other resamples
    end for
    // Intrinsic reward (episode buffer)
    for each robot i **do**
      If max dist to buffer $\geq \tau$: generate $r_i^{t,\text{int}}$, update buffer
    end for
    Execute joint action $(a_1^t, \ldots, a_N^t)$, collect rewards
    Store (o^t, a^t, r^t) in $\mathcal{B}$
  end for
  // PPO update
  Sample mini-batch from $\mathcal{B}$
  Compute advantages via GAE
  Update ϕ, ψ, θ using PPO loss (Eq. 17)
end for

III. EXPERIMENTS

A. Experimental Setup

Training. The number of robots is set to 8 with a 3×3 FOV. Grid world sizes are randomly sampled from $\{10\times10,\ 25\times25,\ 40\times40\}$; obstacle density follows a triangular distribution over $[0\%, 50\%]$ with peak at 33%. A representative training environment is shown in Fig. 5; full parameter settings are listed in Table III.

Testing. The robot count is scaled to 128 on a 40×40 grid at obstacle densities of 0%, 15%, and 30%. This train–test discrepancy evaluates generalization across scales.

Scalability evaluation. To further verify generalization across different robot counts, the model trained with 8 robots is tested with 16, 32, 64, and 128 robots. Fig. 6 shows that while success rate decreases as the number of robots increases, RMHA maintains high success rates from 16 to 64 robots and avoids performance collapse at 128 robots, demonstrating good scalability.

All experiments are conducted on an Ubuntu server with dual NVIDIA RTX 8000 GPUs, using PyTorch 2.1 and Python 3.7.

B. Baselines and Evaluation Metrics

Baselines. We compare with three state-of-the-art MARL-based MRPP methods: **SCRIMP** [1] (GNN-based scalable

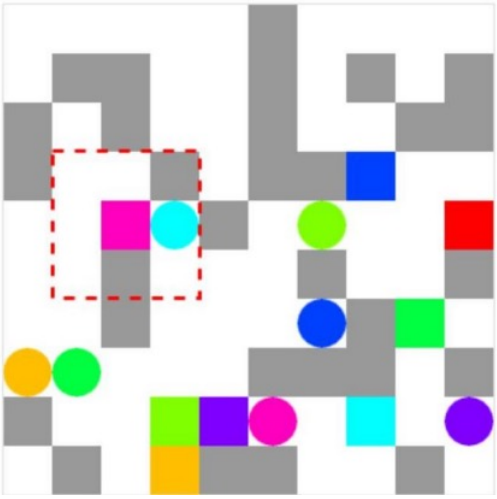


Fig. 5. A 10×10 grid world with 30% obstacle density and 8 robots. Colored squares indicate start positions; colored circles indicate goals; gray circles represent obstacles; the red dashed box shows the 3×3 FOV.

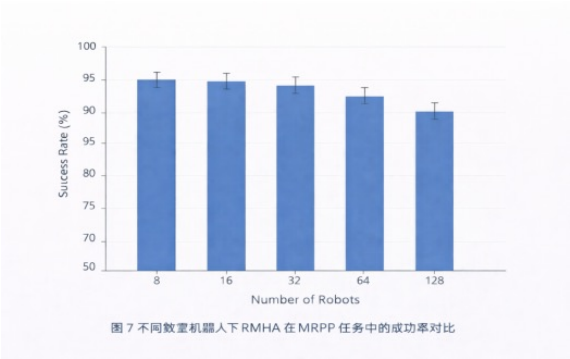


Fig. 6. Success rate of RMHA under different robot counts.

communication, FOV 3×3), **DHC** [17] (GNN-based communication, FOV 9×9), and **PICO** (ad-hoc routing communication, FOV 11×11). We also include the classical centralized planner **ODrM*** [10] with inflation factor $\epsilon = 2.0$ and a 5-minute timeout.

Metrics.

- **Max Reached (MR):** the maximum number of robots that reach their goals at any time step.
- **Collision Rate (CO):** the ratio of collision events (with obstacles or other robots) to the product of episode length and robot count.
- **Success Rate (SR):** the percentage of episodes in which all robots reach their goals collision-free within the time limit $T_{\max}$:

$$S_r = \frac{N_{\text{success}}}{N_{\text{total}}} \times 100\%. \tag{18}$$

Statistical significance is ensured by running at least 100 episodes per setting.

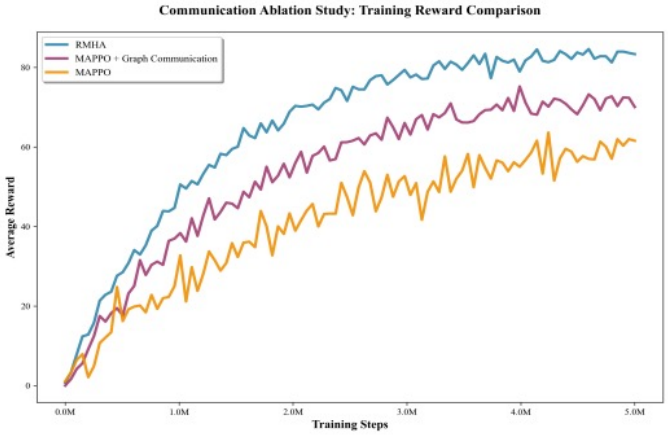
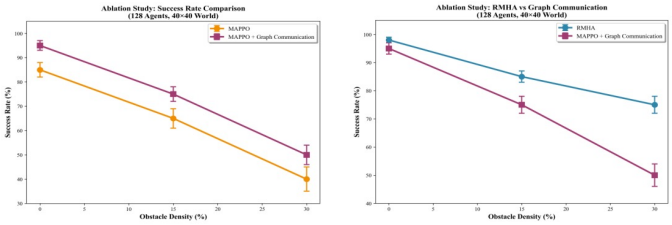


Fig. 7. Communication ablation: training reward curves.



(a) MAPPO vs. Graph Comm (b) RMHA vs. Graph Comm

Fig. 8. Communication ablation: success rate comparison.

C. Results and Analysis

1) Communication Ablation Study: To evaluate the contribution of the Transformer-based communication mechanism, we develop two ablation baselines:

- **MAPPO**: the communication module is entirely removed; robots never send or receive messages.
- **MAPPO+Graph Comm**: attention weights are computed based on message content alone, without fusing Manhattan distance embeddings.

All experiments use 128 robots in a 40×40 grid at obstacle densities of 0%, 15%, and 30%.

Fig. 7 shows the training reward over time. All three methods exhibit rapid reward increase in the early stages. However, RMHA and MAPPO+Graph Comm converge faster and more stably, while MAPPO rises more slowly with larger variance, indicating the significant advantages of graph communication in training efficiency and convergence.

As shown in Fig. 8, success rates decrease with increasing obstacle density. MAPPO+Graph Comm consistently outperforms MAPPO—at 30% density, MAPPO drops to approximately 40% while Graph Comm remains around 50%. RMHA further improves upon Graph Comm, achieving approximately 75% at 30% density (vs. 50%), confirming that Manhattan distance-based relation encoding enables more accurate spatial understanding. In terms of maximum goals reached, RMHA maintains near 125 goals even in high-density environments, while MAPPO degrades rapidly.

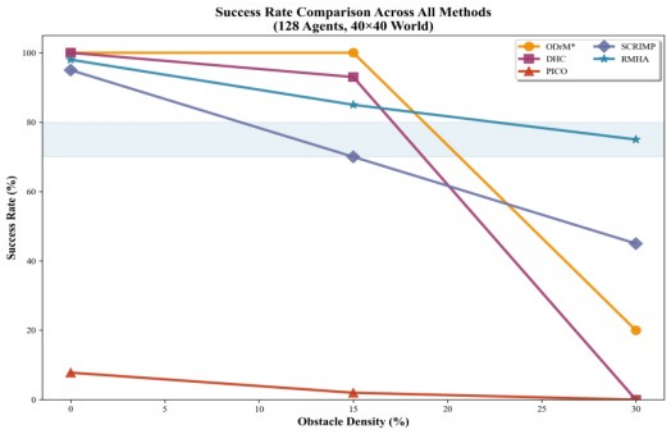


Fig. 9. Success rate comparison across algorithms on Warehouse and City/Game maps.

Overall, in the 128-robot, 30%-density setting, the success rate of RMHA exceeds that of the non-enhanced communication method by 53%, verifying the algorithm's generalization capability and robustness at scale.

2) Comparison with State-of-the-Art Methods: To further evaluate RMHA, we compare it with ODrM*, SCRIMP, DHC, and PICO on two types of benchmark maps: **Warehouse** maps (regular shelf aisles and narrow corridors simulating pick-and-place scenarios) and **City/Game** maps (irregular block/maze-like layouts simulating urban traffic or game levels). All experiments use a 40×40 grid with 128 robots. For each map type and obstacle complexity level (0%, 15%, 30%), 100 reproducible map instances are sampled with fixed random seeds; all algorithms run on the same instances to ensure fairness. Mean values with confidence intervals are reported.

As shown in Fig. 9, all algorithms' success rates decline with increasing obstacle complexity. At low complexity (0%), RMHA performs comparably to ODrM* and SCRIMP. As complexity increases to 15% and 30%, RMHA's advantage becomes progressively larger. At 30%, RMHA maintains a success rate of approximately 75%, while SCRIMP falls below 50%, ODrM* drops to roughly 20% (within the 5-minute planning limit), and DHC and PICO exhibit severe performance degradation.

Regarding maximum goals reached, PICO and DHC degrade fastest despite having communication mechanisms. In contrast, RMHA maintains 120+ goals even at 30% density, benefiting from spatial relation encoding for more precise modeling of local congestion structure.

3) Trajectory Visualization: Fig. 10 compares the global trajectories of RMHA and MAPPO in a complex environment. RMHA produces well-organized trajectories with fewer path conflicts, whereas MAPPO exhibits significant crossing and clustering in local areas, with some robots stalling for extended periods.

Fig. 11 shows a representative head-on encounter in a narrow corridor. Under the baseline, both robots adopt similar forward/wait strategies, resulting in persistent deadlock. Under

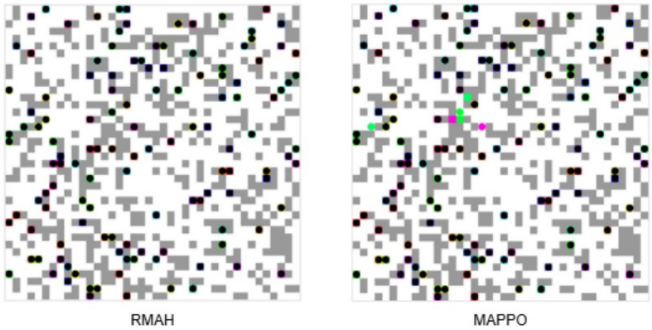


Fig. 10. Global trajectory comparison: RMHA (left) vs. MAPPO (right).

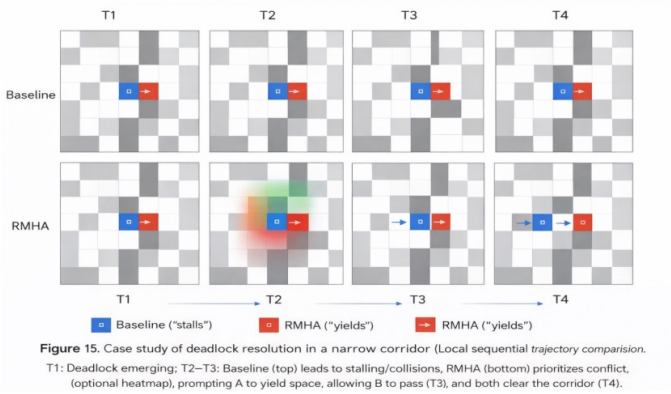


Fig. 11. Deadlock resolution in a narrow corridor. At T_1, robots A and B enter a conflict; the baseline leads to deadlock by T_2–T_3. Under RMHA, robot A yields at T_3, allowing B to pass, and both proceed by T_4.

RMHA, the attention mechanism enables robot A to perceive the conflicting neighbor's proximity and proactively yield, resolving the deadlock within two time steps.

IV. Conclusion

This paper investigated the cooperative communication problem in multi-robot path planning from the perspectives of communication mechanism modeling, algorithmic framework integration, and experimental validation. We proposed RMHA, a spatial relation-enhanced multi-head attention communication method based on graph attention mechanisms. By incorporating relative distance information between robots into the attention weight computation, RMHA dynamically adjusts communication weights, significantly reducing communication overhead while improving path planning success rates.

Experiments demonstrate that RMHA exhibits high adaptability and robustness across varying obstacle densities, with particularly strong performance in high-density scenarios where its success rate substantially exceeds existing methods. Communication ablation studies further confirm the critical role of inter-robot distance information in optimizing path planning decisions.

In summary, RMHA provides an effective solution for efficient cooperation in complex multi-robot environments,

offering a new perspective on intelligent agent communication and collaboration.

Limitations and future work. The current evaluation is limited to discrete grid worlds with uniform robot capabilities; extending RMHA to continuous action spaces and heterogeneous robot teams remains open. Additionally, the Manhattan distance embedding assumes obstacle-free line-of-sight proximity, which may not reflect true reachability in cluttered environments. Future work will explore learned distance metrics that account for obstacle topology, as well as integration with lifelong MRPP settings where start–goal assignments change dynamically.

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